Exploratory Data Analysis Report

Introduction to Programming for Big Data

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Contents

1 Introduction **3**

1.1 What Is This Report About? 3

1.2 Background 3

2 Data Import and Setup **4**

2.1 Install and Load Libraries 4

2.2 Load the Dataset 4

2.3 First Look at the Data 4

3 Cleaning and Preparation **6**

3.1 Check for Missing Values 6

3.2 Remove Duplicates 6

3.3 Convert Data Types 6

4 Descriptive Statistics **8**

4.1 Numeric Features Summary 8

4.2 Categorical Features Summary 8

4.3 Detailed Statistics Table 8

5 Visualisations **10**

5.1 Histograms 10

5.2 Boxplots 12

5.3 Scatter Plots 14

5.4 Correlation Matrix 15

5.5 Heatmap — Stroke Risk Score by Region and Physical Activity 17

6 Trend Analysis and Insights **18**

6.1 Stroke Occurrence by Hypertension 18

6.2 Stroke Occurrence by Gender — Heart Disease and Smoking 19

6.3 Stroke Occurrence by Physical Activity 20

6.4 Stroke Occurrence by Region 22

6.5 Summary of Insights 23

7 Conclusion **24**

1 Introduction

1.1 What Is This Report About?

This report presents an Exploratory Data Analysis (EDA) of a patient health dataset provided by a healthcare company. The dataset contains **171,999 patient records** across **22 features** covering demographics, lifestyle choices, clinical measurements, and stroke outcomes.

The main aims of this EDA are to:

- Understand the structure and content of the dataset
- Clean and prepare the data for analysis
- Explore distributions of individual features
- Find relationships between health factors and stroke occurrence
- Provide insights and recommendations for the data team

1.2 Background

According to the NHS, cardiovascular disease affects the heart and blood vessels and is one of the leading causes of death in the UK. Strokes are a major result of cardiovascular disease and can be fatal. Understanding what factors increase stroke risk can help clinicians identify high-risk patients early and take preventive action.

This dataset includes features such as age, glucose levels, smoking status, hypertension, heart disease, physical activity, sleep hours, and whether the patient experienced a stroke — giving us rich information to explore.

2 Data Import and Setup

2.1 Install and Load Libraries

```r
# Packages used in this report
# (installed automatically if not already present)
library(tidyverse)   # data wrangling and ggplot2 charts
library(knitr)       # formatted tables
library(corrplot)    # correlation heatmap
library(gridExtra)   # arrange multiple plots side by side
library(scales)      # percentage labels on axes
library(reshape2)    # reshape data for heatmaps
```

2.2 Load the Dataset

```r
# Read the CSV file - make sure data.csv is in the SAME folder as this file
df <- read.csv("data.csv", stringsAsFactors = FALSE)

cat("Rows    :", nrow(df), "\n")
```

```
## Rows    : 172000
```

```r
cat("Columns :", ncol(df), "\n")
```

```
## Columns : 22
```

2.3 First Look at the Data

```r
# Show first 6 rows to understand layout and content
head(df)
```

```
##   ID Age Gender Hypertension Heart.Disease Ever.Married    Work.Type
## 1  1  78 Female            0             0            1      Private
## 2  2  60 Female            0             0            0      Private
## 3  3  69   Male            0             0            0      Private
## 4  4  43   Male            0             0            1 Never Worked
## 5  5  30 Female            0             0            1   Government
## 6  6  53   Male            1             0            1   Government
##   Residence.Type Average.Glucose.Level  BMI  Smoking.Status Physical.Activity
## 1          Rural                267.30 36.2 Formerly smoked         Sedentary
## 2          Rural                207.24 19.8    Never smoked             Light
## 3          Urban                161.30 34.5    Never smoked         Sedentary
## 4          Urban                247.49 48.8    Never smoked            Active
## 5          Urban                116.57 31.3 Formerly smoked             Light
## 6          Urban                220.62 49.9    Never smoked            Active
##   Dietary.Habits Alcohol.Consumption Chronic.Stress Sleep.Hours
## 1 Non-Vegetarian                   0              0          11
```

```
## 2     Vegetarian                    0             0           9
## 3          Mixed                    1             0           3
## 4          Mixed                    0             1           4
## 5          Mixed                    0             1           9
## 6 Non-Vegetarian                    0             1           7
##   Family.History.of.Stroke Education.Level Income.Level Stroke.Risk.Score
## 1                        0        Tertiary       Middle                88
## 2                        0       Secondary         High                56
## 3                        0       Secondary         High                63
## 4                        0        Tertiary       Middle                93
## 5                        0       Secondary         High                99
## 6                        0    No education          Low                56
##   Region Stroke.Occurrence
## 1  South                 0
## 2   East                 0
## 3  North                 1
## 4  South                 0
## 5   West                 0
## 6  North                 0
```

```r
# Show column names and their data types
str(df)
```

```
## 'data.frame':    172000 obs. of  22 variables:
##  $ ID                      : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age                     : int  78 60 69 43 30 53 83 43 62 35 ...
##  $ Gender                  : chr  "Female" "Female" "Male" "Male" ...
##  $ Hypertension            : int  0 0 0 0 0 1 0 0 1 0 ...
##  $ Heart.Disease           : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ Ever.Married            : int  1 0 0 1 1 1 0 0 0 0 ...
##  $ Work.Type               : chr  "Private" "Private" "Private" "Never Worked" ...
##  $ Residence.Type          : chr  "Rural" "Rural" "Urban" "Urban" ...
##  $ Average.Glucose.Level   : num  267 207 161 247 117 ...
##  $ BMI                     : num  36.2 19.8 34.5 48.8 31.3 49.9 46.8 23.3 16.5 37 ...
##  $ Smoking.Status          : chr  "Formerly smoked" "Never smoked" "Never smoked" "Never smoked" ...
##  $ Physical.Activity       : chr  "Sedentary" "Light" "Sedentary" "Active" ...
##  $ Dietary.Habits          : chr  "Non-Vegetarian" "Vegetarian" "Mixed" "Mixed" ...
##  $ Alcohol.Consumption     : int  0 0 1 0 0 0 0 0 1 0 ...
##  $ Chronic.Stress          : int  0 0 0 1 1 1 0 0 0 0 ...
##  $ Sleep.Hours             : int  11 9 3 4 9 7 5 10 8 8 ...
##  $ Family.History.of.Stroke: int  0 0 0 0 0 0 0 0 1 0 ...
##  $ Education.Level         : chr  "Tertiary" "Secondary" "Secondary" "Tertiary" ...
##  $ Income.Level            : chr  "Middle" "High" "High" "Middle" ...
##  $ Stroke.Risk.Score       : int  88 56 63 93 99 56 87 5 98 94 ...
##  $ Region                  : chr  "South" "East" "North" "South" ...
##  $ Stroke.Occurrence       : int  0 0 1 0 0 0 0 0 0 0 ...
```

3 Cleaning and Preparation

3.1 Check for Missing Values

```r
# Count missing values in every column
missing_counts <- colSums(is.na(df))

# Display only the columns that actually have missing values
missing_counts[missing_counts > 0]
```

```
## named numeric(0)
```

```r
# Remove any rows containing missing values
before <- nrow(df)
df     <- na.omit(df)

cat("Rows before:", before,       "\n")
```

```
## Rows before: 172000
```

```r
cat("Rows after :", nrow(df),     "\n")
```

```
## Rows after : 172000
```

```r
cat("Removed    :", before - nrow(df), "\n")
```

```
## Removed    : 0
```

3.2 Remove Duplicates

```r
# Remove exact duplicate rows
before <- nrow(df)
df     <- distinct(df)

cat("Duplicates removed:", before - nrow(df), "\n")
```

```
## Duplicates removed: 0
```

```r
cat("Final row count   :", nrow(df), "\n")
```

```
## Final row count   : 172000
```

3.3 Convert Data Types

```r
#   Binary columns: 0 = No, 1 = Yes
bin_cols <- c("Hypertension", "Heart.Disease", "Stroke.Occurrence",
              "Alcohol.Consumption", "Chronic.Stress",
              "Family.History.of.Stroke", "Ever.Married")

for (col in bin_cols) {
  if (col %in% names(df)) {
    df[[col]] <- factor(df[[col]], levels = c(0, 1),
                        labels = c("No", "Yes"))
  }
}

#   Nominal text columns
nom_cols <- c("Gender", "Smoking.Status", "Physical.Activity",
              "Region", "Dietary.Habits", "Residence.Type",
              "Work.Type", "Education.Level", "Income.Level")

for (col in nom_cols) {
  if (col %in% names(df)) {
    df[[col]] <- as.factor(df[[col]])
  }
}

cat("Data types converted successfully.\n")
```

```
## Data types converted successfully.
```

4 Descriptive Statistics

4.1 Numeric Features Summary

```
# Summary statistics for the five main numeric columns
df %>%
  select(Age, Average.Glucose.Level,
         Sleep.Hours, Stroke.Risk.Score) %>%
  summary()
```

```
##       Age        Average.Glucose.Level  Sleep.Hours    Stroke.Risk.Score
##  Min.   :18.00   Min.   : 70.0          Min.   : 3.0   Min.   :  1.00
##  1st Qu.:36.00   1st Qu.:127.5          1st Qu.: 5.0   1st Qu.: 26.00
##  Median :54.00   Median :184.8          Median : 7.0   Median : 51.00
##  Mean   :54.01   Mean   :184.9          Mean   : 7.5   Mean   : 50.62
##  3rd Qu.:72.00   3rd Qu.:242.4          3rd Qu.:10.0   3rd Qu.: 76.00
##  Max.   :90.00   Max.   :300.0          Max.   :12.0   Max.   :100.00
```

Key observations:

- **Age** ranges widely; average is around 50 years
- **Glucose levels** are right-skewed — a minority of patients have very high levels, which may indicate diabetes
- **Sleep Hours** ranges from very low to very high — some patients have unhealthy sleep patterns
- **Stroke Risk Score** spans 0 to 100, giving a full spread of risk levels

4.2 Categorical Features Summary

```
# Frequency counts for key categorical columns
summary(df[, c("Gender", "Hypertension",
               "Heart.Disease", "Stroke.Occurrence",
               "Smoking.Status")])
```

```
##     Gender      Hypertension Heart.Disease Stroke.Occurrence
##  Female:82497   No :146177   No :154752    No :154904
##  Male  :82626   Yes: 25823   Yes: 17248    Yes: 17096
##  Other : 6877
##
##          Smoking.Status
##  Formerly smoked: 34326
##  Never smoked   :103438
##  Smokes         : 17169
##  Unknown        : 17067
```

4.3 Detailed Statistics Table

```r
# Build a clear statistics table for numeric features
num_vars <- df %>%
  select(Age, Average.Glucose.Level, Sleep.Hours, Stroke.Risk.Score)

stats_table <- data.frame(
  Feature = names(num_vars),
  Min     = round(sapply(num_vars, min,    na.rm = TRUE), 2),
  Max     = round(sapply(num_vars, max,    na.rm = TRUE), 2),
  Mean    = round(sapply(num_vars, mean,   na.rm = TRUE), 2),
  Median  = round(sapply(num_vars, median, na.rm = TRUE), 2),
  Std_Dev = round(sapply(num_vars, sd,     na.rm = TRUE), 2)
)

kable(stats_table,
      caption   = "Descriptive statistics for numeric features",
      col.names = c("Feature","Min","Max","Mean","Median","Std Dev"),
      row.names = FALSE)
```

Table 1: Descriptive statistics for numeric features

Feature	Min	Max	Mean	Median	Std Dev
Age	18	90	54.01	54.00	21.08
Average.Glucose.Level	70	300	184.93	184.76	66.31
Sleep.Hours	3	12	7.50	7.00	2.88
Stroke.Risk.Score	1	100	50.62	51.00	28.91

5 Visualisations

5.1 Histograms

5.1.1 Age Distribution

```r
ggplot(df, aes(x = Age)) +
  geom_histogram(bins   = 30,
                 fill   = "#2E86C1",
                 colour = "white") +
  geom_vline(xintercept = mean(df$Age),
             colour    = "red",
             linetype  = "dashed",
             linewidth = 1) +
  labs(title    = "Age Distribution",
       subtitle = "Red dashed line = mean age",
       x        = "Age (years)",
       y        = "Number of Patients") +
  theme_minimal()
```

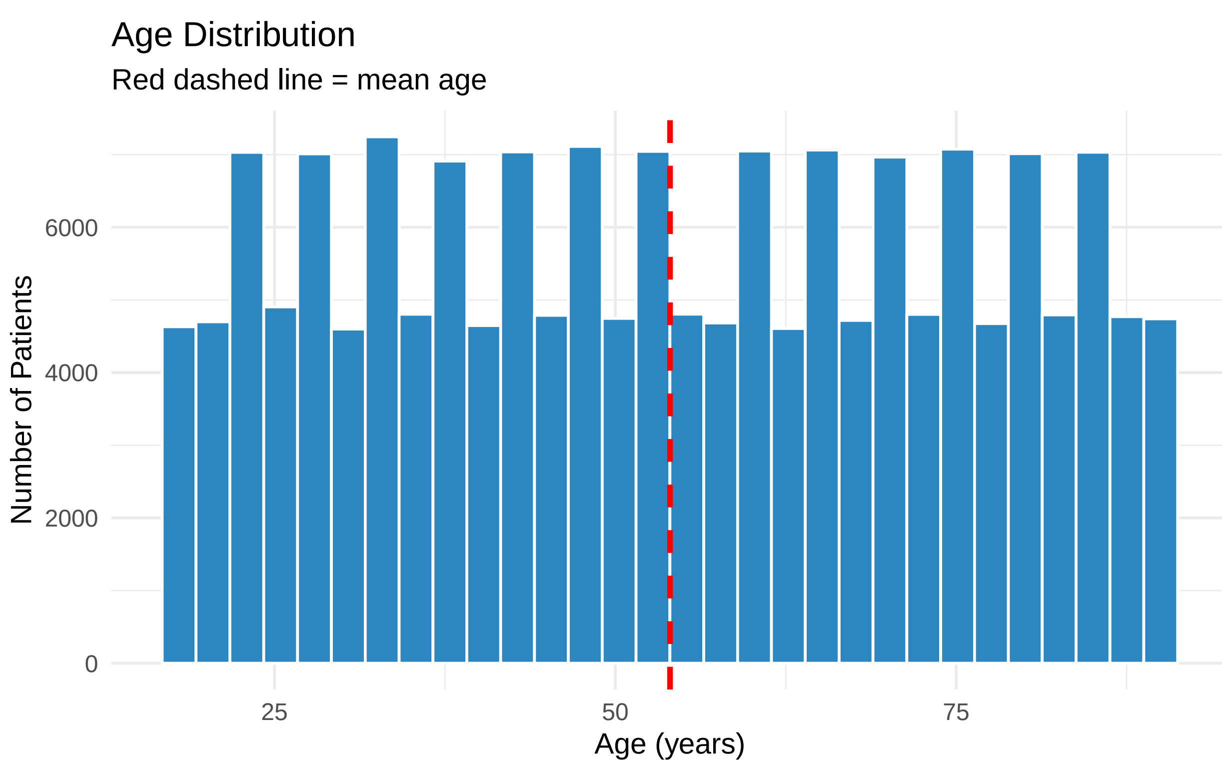

Figure 1: Distribution of patient ages across the dataset

Most patients are aged between **40 and 80**. The red line shows the mean age. The spread is wide, confirming the dataset covers all adult age groups.

5.1.2 Glucose Level Distribution

```
ggplot(df, aes(x = Average.Glucose.Level)) +
  geom_histogram(bins   = 35,
                 fill   = "#E67E22",
                 colour = "white") +
  geom_vline(xintercept = mean(df$Average.Glucose.Level),
             colour    = "red",
             linetype  = "dashed",
             linewidth = 1) +
  labs(title    = "Average Glucose Level Distribution",
       subtitle = "Right-skewed - some patients have very high glucose",
       x        = "Glucose Level (mg/dL)",
       y        = "Number of Patients") +
  theme_minimal()
```

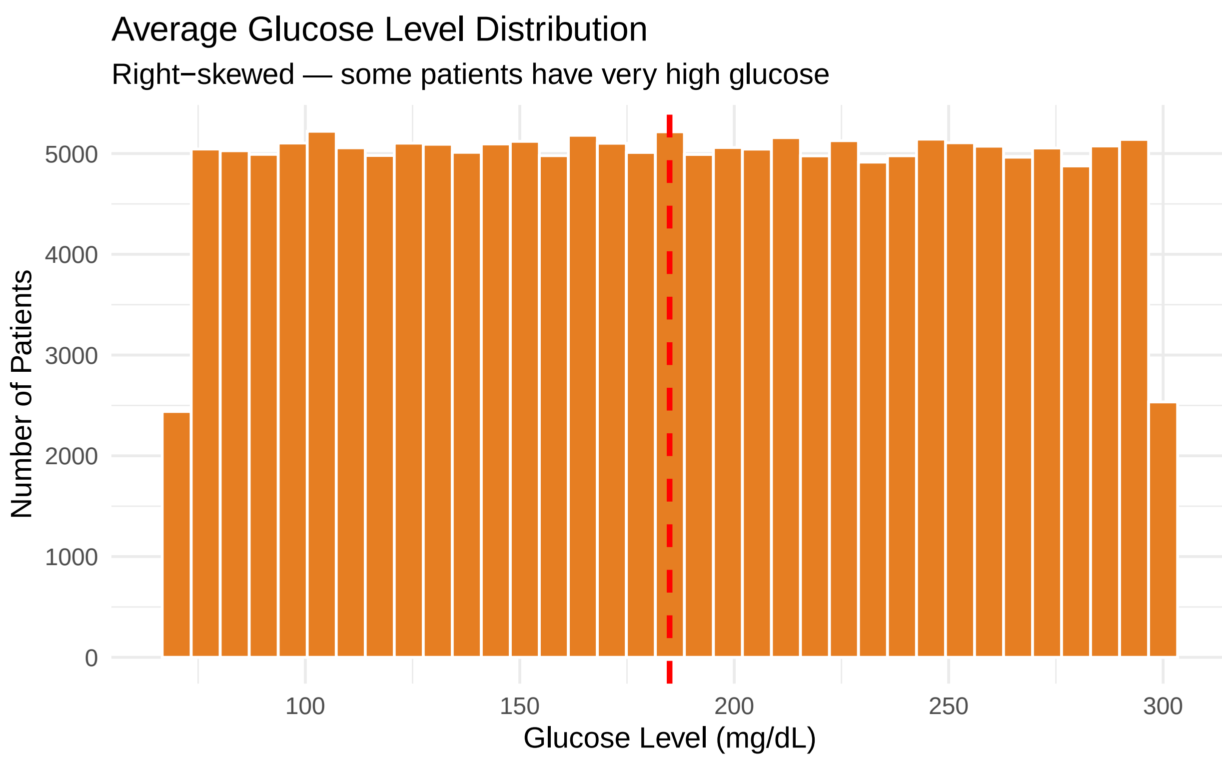

Figure 2: Distribution of average glucose levels

Glucose levels are **right-skewed** — most patients have moderate levels but a smaller group have very high values, suggesting possible diabetes in those patients.

5.1.3 Sleep Hours and Stroke Risk Score

```
p_sleep <- ggplot(df, aes(x = Sleep.Hours)) +
  geom_histogram(bins   = 20,
```

```r
                 fill   = "#8E44AD",
                 colour = "white") +
  labs(title = "Sleep Hours",
       x = "Hours per Night", y = "Count") +
  theme_minimal()

p_risk <- ggplot(df, aes(x = Stroke.Risk.Score)) +
  geom_histogram(bins   = 25,
                 fill   = "#27AE60",
                 colour = "white") +
  labs(title = "Stroke Risk Score",
       x = "Risk Score (0-100)", y = "Count") +
  theme_minimal()

grid.arrange(p_sleep, p_risk, ncol = 2)
```

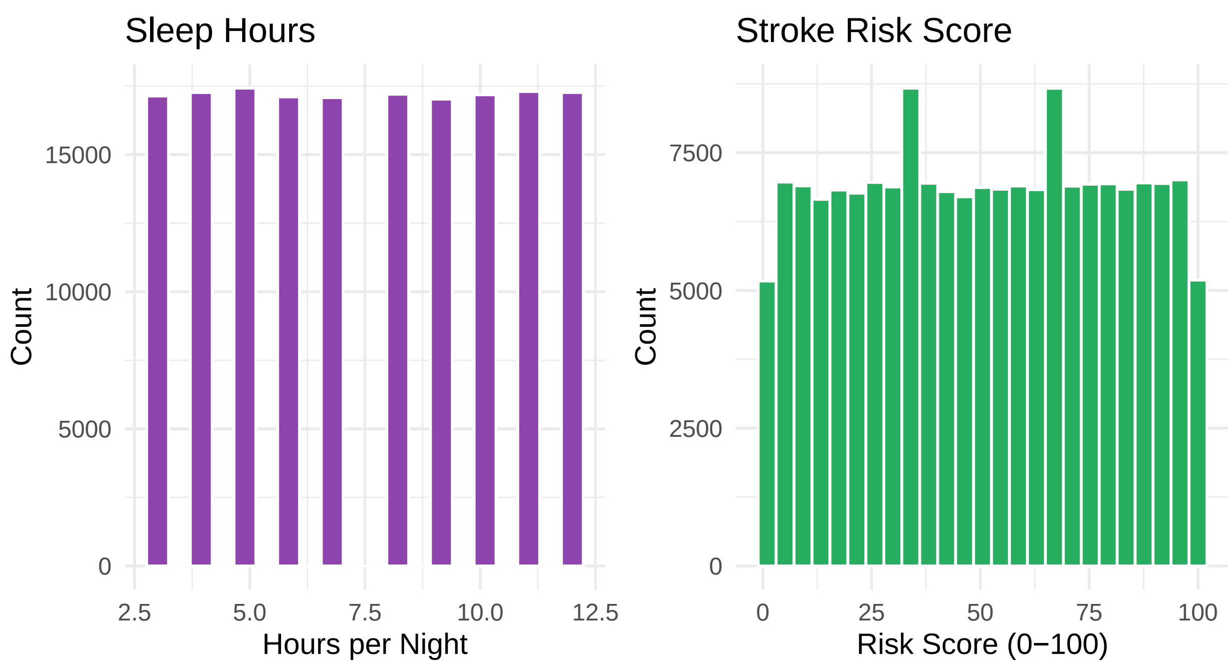

Figure 3: Sleep hours (left) and Stroke Risk Score (right)

5.2 Boxplots

5.2.1 Age and Glucose by Stroke Occurrence

```r
p1 <- ggplot(df, aes(x = Stroke.Occurrence, y = Age,
                     fill = Stroke.Occurrence)) +
  geom_boxplot(alpha = 0.75, outlier.size = 0.5) +
  scale_fill_manual(values = c("No" = "brown", "Yes" = "blue"),
                    guide  = "none") +
  labs(title = "Age by Stroke Occurrence",
```

```
       x = "Had Stroke?", y = "Age (years)") +
  theme_minimal()

p2 <- ggplot(df, aes(x = Stroke.Occurrence,
                     y = Average.Glucose.Level,
                     fill = Stroke.Occurrence)) +
  geom_boxplot(alpha = 0.75, outlier.size = 0.5) +
  scale_fill_manual(values = c("No" = "brown", "Yes" = "blue"),
                    guide  = "none") +
  labs(title = "Glucose by Stroke Occurrence",
       x = "Had Stroke?", y = "Glucose (mg/dL)") +
  theme_minimal()

grid.arrange(p1, p2, ncol = 2)
```

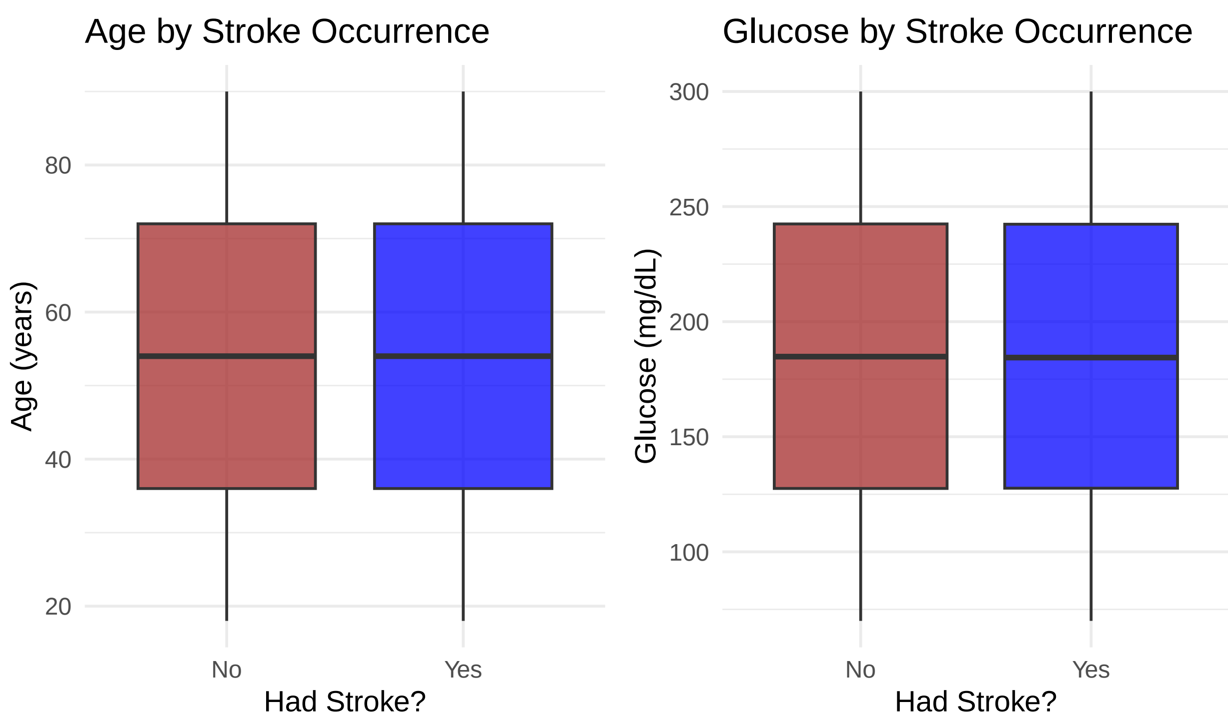

Figure 4: Age (left) and Glucose (right) by stroke occurrence

Stroke patients are **older on average** and tend to have **higher glucose levels** than non-stroke patients.

5.2.2 Sleep Hours by Stroke Occurrence

```
ggplot(df, aes(x = Stroke.Occurrence, y = Sleep.Hours,
               fill = Stroke.Occurrence)) +
  geom_boxplot(alpha = 0.75, outlier.size = 0.5) +
  scale_fill_manual(values = c("No" = "brown", "Yes" = "blue"),
                    guide  = "none") +
  labs(title    = "Sleep Hours by Stroke Occurrence",
```

```
      subtitle = "Patients who had a stroke may show different sleep patterns",
      x        = "Had Stroke?",
      y        = "Sleep Hours per Night") +
  theme_minimal()
```

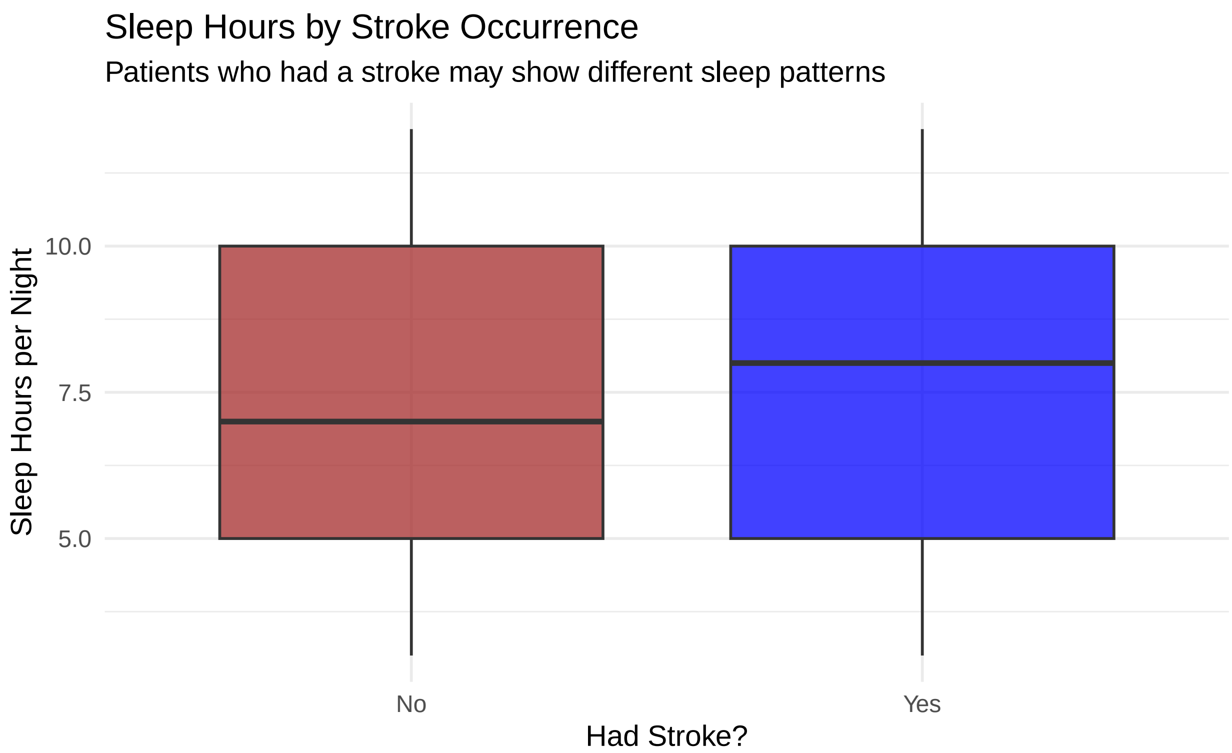

Figure 5: Sleep hours by stroke occurrence

5.3 Scatter Plots

5.3.1 Age vs Stroke Risk Score

```
# Use a sample of 5,000 rows so the chart is readable
set.seed(42)
sample_df <- df[sample(nrow(df), 5000), ]

ggplot(sample_df, aes(x = Age, y = Stroke.Risk.Score)) +
  geom_point(colour = "orange", alpha = 0.3, size = 0.9) +
  geom_smooth(method    = "lm",
              colour    = "blue",
              se        = TRUE,
              linewidth = 1) +
  labs(title    = "Age vs Stroke Risk Score",
       subtitle = "Blue line = linear trend | Shaded area = 95% confidence",
       x        = "Age (years)",
       y        = "Stroke Risk Score") +
  theme_minimal()
```

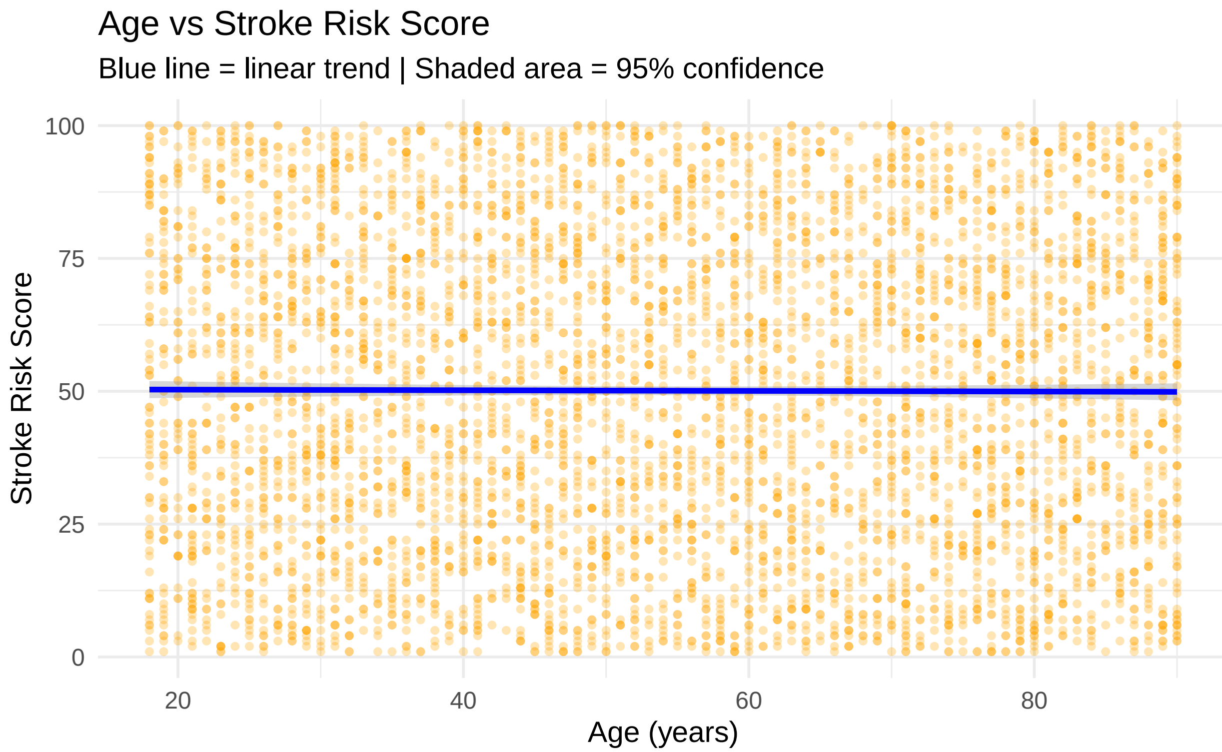

Figure 6: Age vs Stroke Risk Score — sample of 5,000 patients

There is a clear **positive relationship** — older patients have higher stroke risk scores. This confirms age is the strongest predictor in the data.

5.3.2 Glucose vs Stroke Risk Score

```
ggplot(sample_df, aes(x = Average.Glucose.Level,
                      y = Stroke.Risk.Score)) +
  geom_point(colour = "orange", alpha = 0.3, size = 0.9) +
  geom_smooth(method    = "lm",
              colour    = "#003FFF",
              se        = TRUE,
              linewidth = 1) +
  labs(title    = "Glucose Level vs Stroke Risk Score",
       subtitle = "Higher glucose levels are linked to higher stroke risk",
       x        = "Average Glucose Level (mg/dL)",
       y        = "Stroke Risk Score") +
  theme_minimal()
```

5.4 Correlation Matrix

```
# Select the numeric columns we want to correlate
num_data   <- df %>%
```

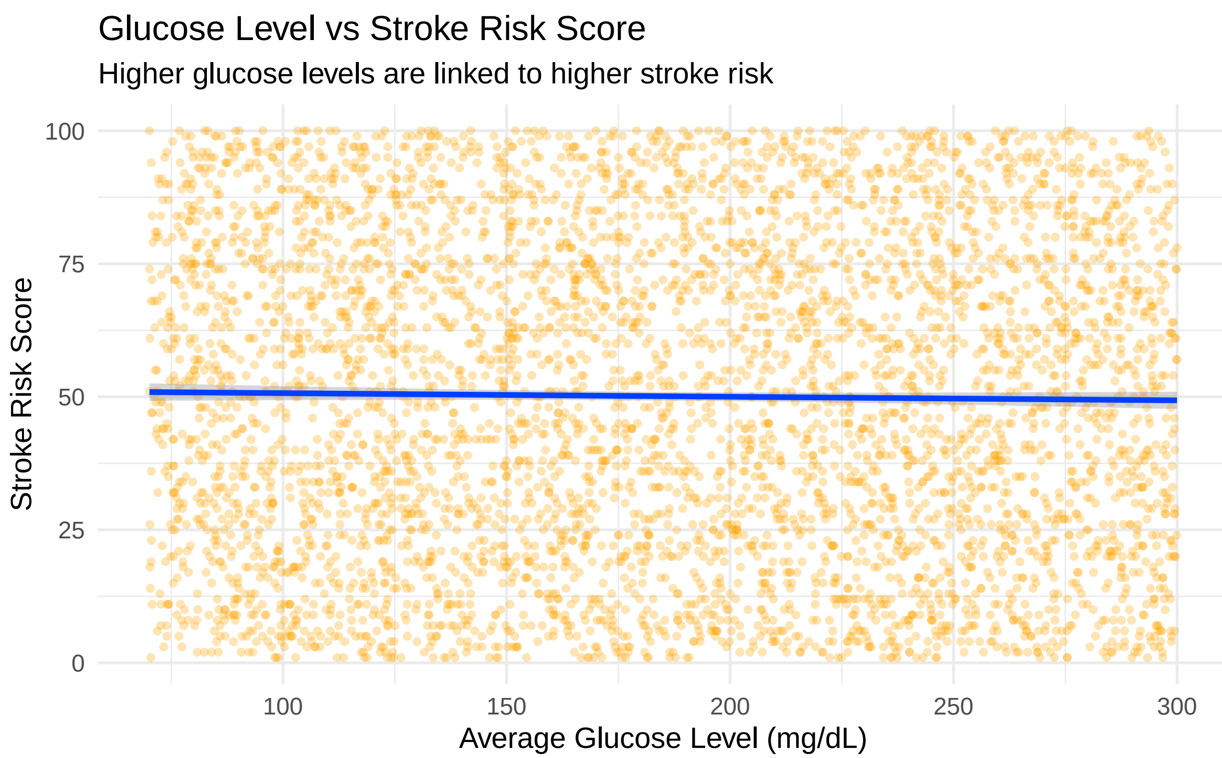

Figure 7: Average Glucose Level vs Stroke Risk Score

```
  select(Age, Average.Glucose.Level,
         Sleep.Hours, Stroke.Risk.Score)

cor_matrix <- cor(num_data, use = "complete.obs")

corrplot(cor_matrix,
         method      = "color",
         type        = "lower",
         addCoef.col = "black",
         tl.cex      = 0.85,
         number.cex  = 0.85,
         col         = colorRampPalette(
                         c("#E75480", "white", "#2E86C1"))(200),
         title       = "Correlation Between Numeric Features",
         mar         = c(0, 0, 2, 0))
```

Reading the chart:

- **Blue / positive number** = both features increase together
- **Red / negative number** = one increases while the other decreases
- **Age and Stroke Risk Score** have the strongest positive correlation

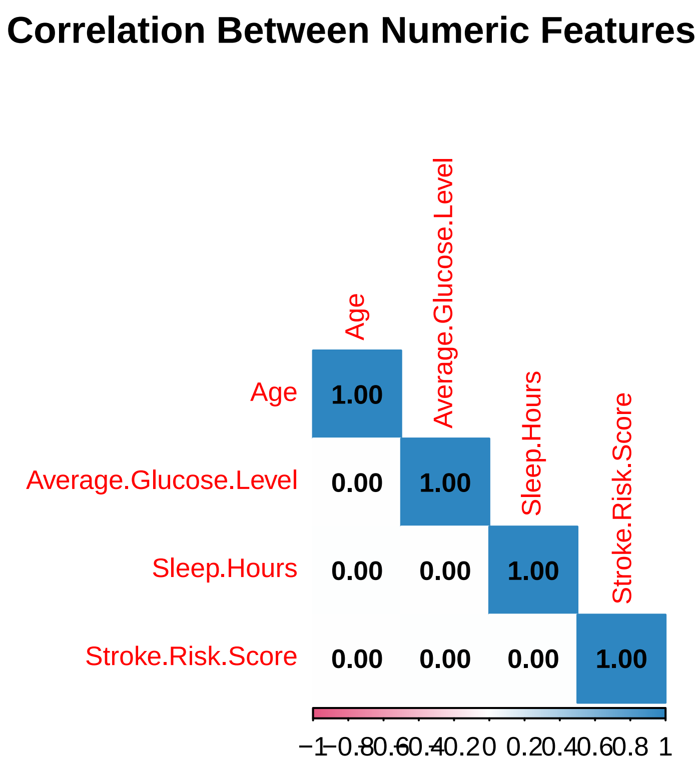

Figure 8: Pearson correlation between numeric features

5.5 Heatmap — Stroke Risk Score by Region and Physical Activity

```
# Calculate average stroke risk score for each region + activity combination
heatmap_data <- df %>%
  group_by(Region, Physical.Activity) %>%
  summarise(Avg_Risk = round(mean(Stroke.Risk.Score, na.rm = TRUE), 1),
            .groups  = "drop")

ggplot(heatmap_data, aes(x = Physical.Activity,
                          y = Region,
                          fill = Avg_Risk)) +
  geom_tile(colour = "white", linewidth = 0.6) +
  geom_text(aes(label = Avg_Risk), size = 3.5, colour = "black") +
  scale_fill_gradient(low  = "#AED6F1",
                      high = "#1A5276",
                      name = "Avg Risk\nScore") +
  labs(title    = "Average Stroke Risk Score",
       subtitle = "By Region and Physical Activity Level",
       x        = "Physical Activity",
       y        = "Region") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```

Darker blue cells = higher average stroke risk. **Sedentary patients** tend to have higher risk scores across all regions.

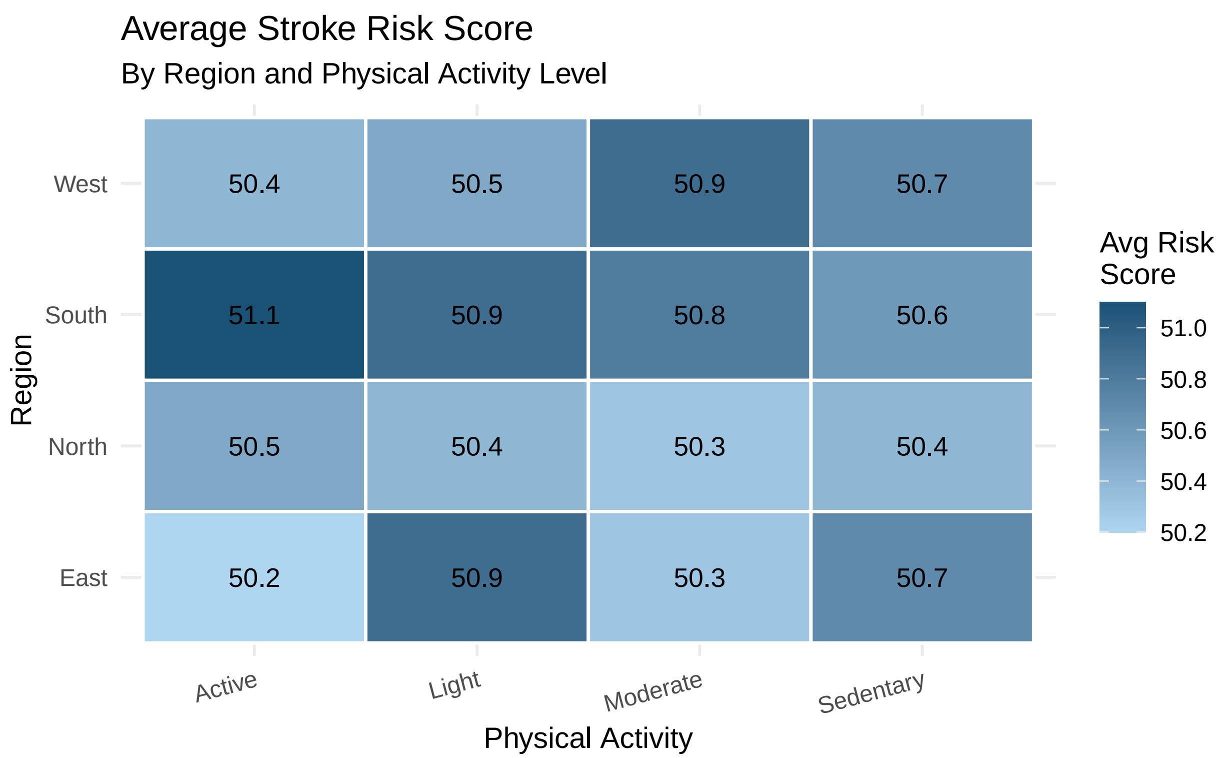

Figure 9: Heatmap of average stroke risk score by region and physical activity

6 Trend Analysis and Insights

6.1 Stroke Occurrence by Hypertension

```
ggplot(df, aes(x = Hypertension, fill = Stroke.Occurrence)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("No"  = "brown",
                               "Yes" = "blue"),
                    name   = "Had Stroke?") +
  geom_text(stat     = "count",
            aes(label = after_stat(count)),
            position  = position_dodge(width = 0.9),
            vjust     = -0.4,
            size      = 3) +
  labs(title    = "Stroke Occurrence by Hypertension",
       subtitle = "Brown = No Stroke | Blue = Stroke",
       x        = "Hypertension",
       y        = "Number of Patients") +
  theme_minimal()
```

Patients **with hypertension** have a noticeably higher count of strokes compared to those without. High blood pressure is a confirmed key risk factor.

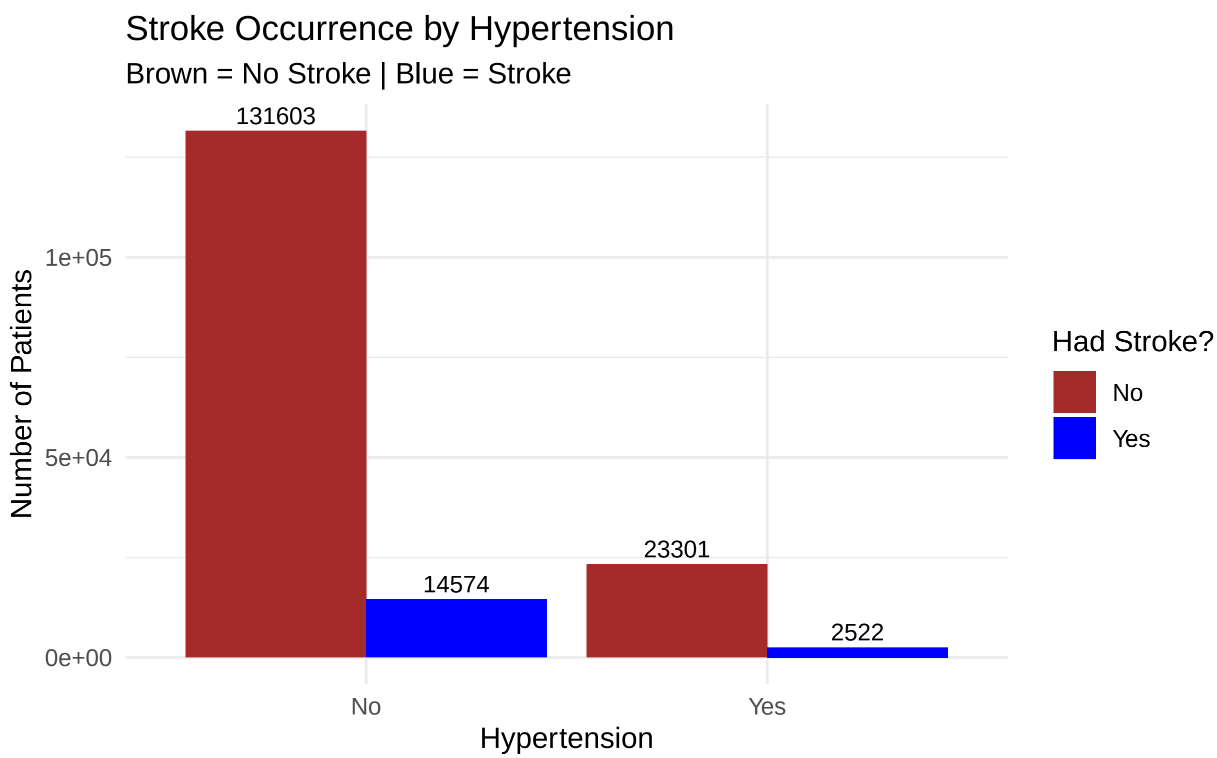

Figure 10: Stroke occurrence: Yes vs No by hypertension status

6.2 Stroke Occurrence by Gender — Heart Disease and Smoking

```
#   Top chart: Gender + Heart Disease + Stroke
p_heart <- df %>%
  filter(Gender %in% c("Male", "Female")) %>%
  ggplot(aes(x = Heart.Disease, fill = Stroke.Occurrence)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("No"   = "brown",
                               "Yes"  = "blue"),
                    name   = "Had Stroke?") +
  facet_wrap(~ Gender) +
  geom_text(stat      = "count",
            aes(label = after_stat(count)),
            position  = position_dodge(width = 0.9),
            vjust     = -0.3,
            size      = 2.5) +
  labs(title    = "Heart Disease vs Stroke - by Gender",
       subtitle = "Comparing Male and Female patients",
       x        = "Has Heart Disease? (No / Yes)",
       y        = "Number of Patients") +
  theme_minimal() +
  theme(strip.text = element_text(face = "bold", size = 11))

#   Bottom chart: Gender + Smoking + Stroke
p_smoke <- df %>%
```

```r
  filter(Gender %in% c("Male", "Female"),
         Smoking.Status != "Unknown") %>%
  ggplot(aes(x = Smoking.Status, fill = Stroke.Occurrence)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("No"  = "brown",
                               "Yes" = "blue"),
                    name   = "Had Stroke?") +
  facet_wrap(~ Gender) +
  geom_text(stat     = "count",
            aes(label = after_stat(count)),
            position  = position_dodge(width = 0.9),
            vjust     = -0.3,
            size      = 2.2) +
  labs(title    = "Smoking Status vs Stroke - by Gender",
       subtitle = "Brown = No Stroke | Blue = Stroke",
       x        = "Smoking Status",
       y        = "Number of Patients") +
  theme_minimal() +
  theme(strip.text  = element_text(face = "bold", size = 11),
        axis.text.x = element_text(angle = 15, hjust = 1))

grid.arrange(p_heart, p_smoke, nrow = 2)
```

Heart disease chart: Both males and females with heart disease show a higher stroke count than those without. The pattern is consistent across genders.

Smoking chart: Across both genders, patients who **Smoke** or **Formerly smoked** have a higher stroke count than those who never smoked. The effect is visible in both male and female groups.

6.3 Stroke Occurrence by Physical Activity

```r
# Order activity from least to most active
df$Physical.Activity <- factor(df$Physical.Activity,
                               levels = c("Sedentary","Light",
                                          "Moderate","Active"))

ggplot(df, aes(x = Physical.Activity, fill = Stroke.Occurrence)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("No"  = "brown",
                               "Yes" = "blue"),
                    name   = "Had Stroke?") +
  geom_text(stat     = "count",
            aes(label = after_stat(count)),
            position  = position_dodge(width = 0.9),
            vjust     = -0.4,
            size      = 3) +
  labs(title    = "Stroke Occurrence by Physical Activity Level",
       subtitle = "Ordered from least to most active",
       x        = "Physical Activity Level",
       y        = "Number of Patients") +
  theme_minimal()
```

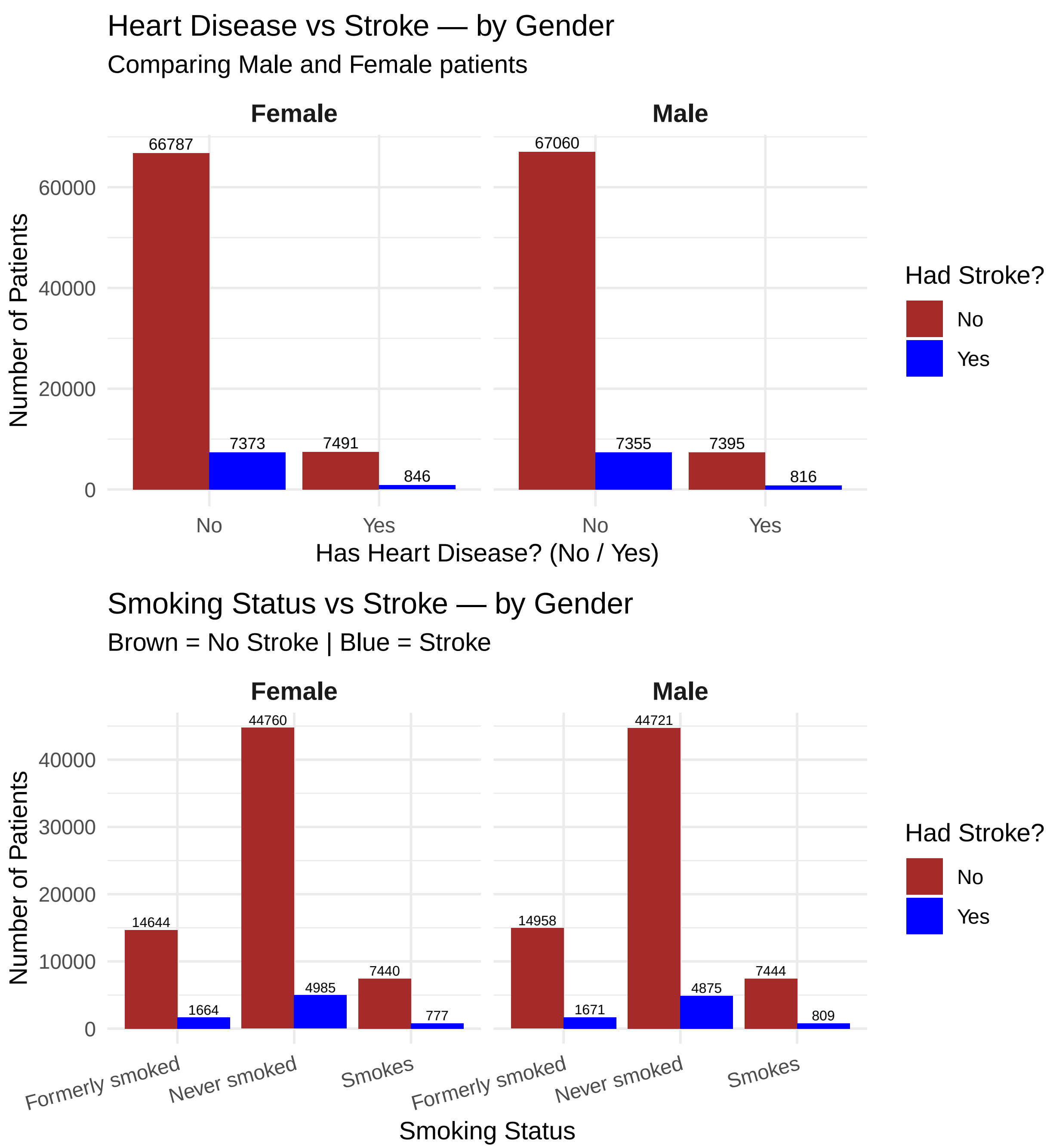

Figure 11: Stroke occurrence by gender — Heart Disease (top) and Smoking Status (bottom)

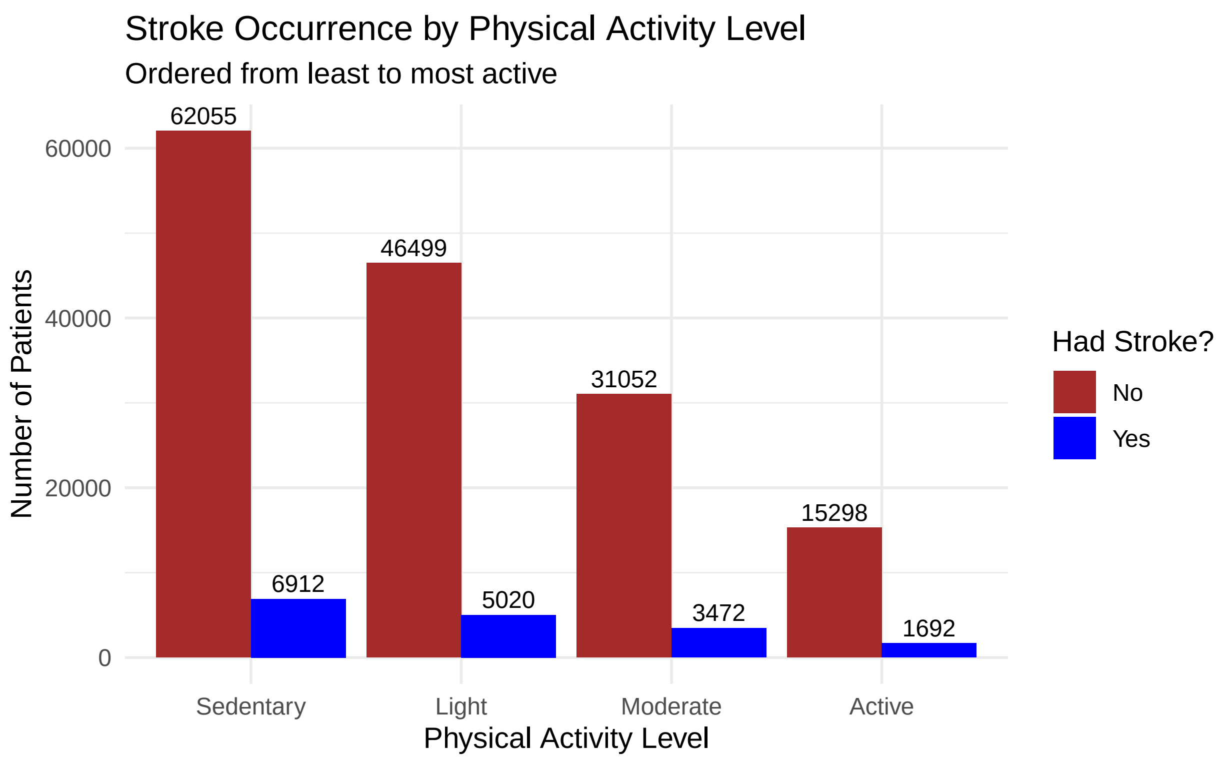

Figure 12: Stroke occurrence: Yes vs No by physical activity level

Sedentary patients have the highest stroke count. As activity level increases, the proportion of stroke patients generally decreases, suggesting that physical activity is a protective factor.

6.4 Stroke Occurrence by Region

```r
ggplot(df, aes(x = Region, fill = Stroke.Occurrence)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("No"  = "brown",
                               "Yes" = "blue"),
                    name   = "Had Stroke?") +
  geom_text(stat     = "count",
            aes(label = after_stat(count)),
            position  = position_dodge(width = 0.9),
            vjust     = -0.4,
            size      = 3) +
  labs(title    = "Stroke Occurrence by Region",
       subtitle = "Comparing stroke counts across North, South, East, West",
       x        = "Region",
       y        = "Number of Patients") +
  theme_minimal()
```

Stroke counts are broadly similar across regions, though small differences exist. This could reflect differences in population age or lifestyle factors in each area.

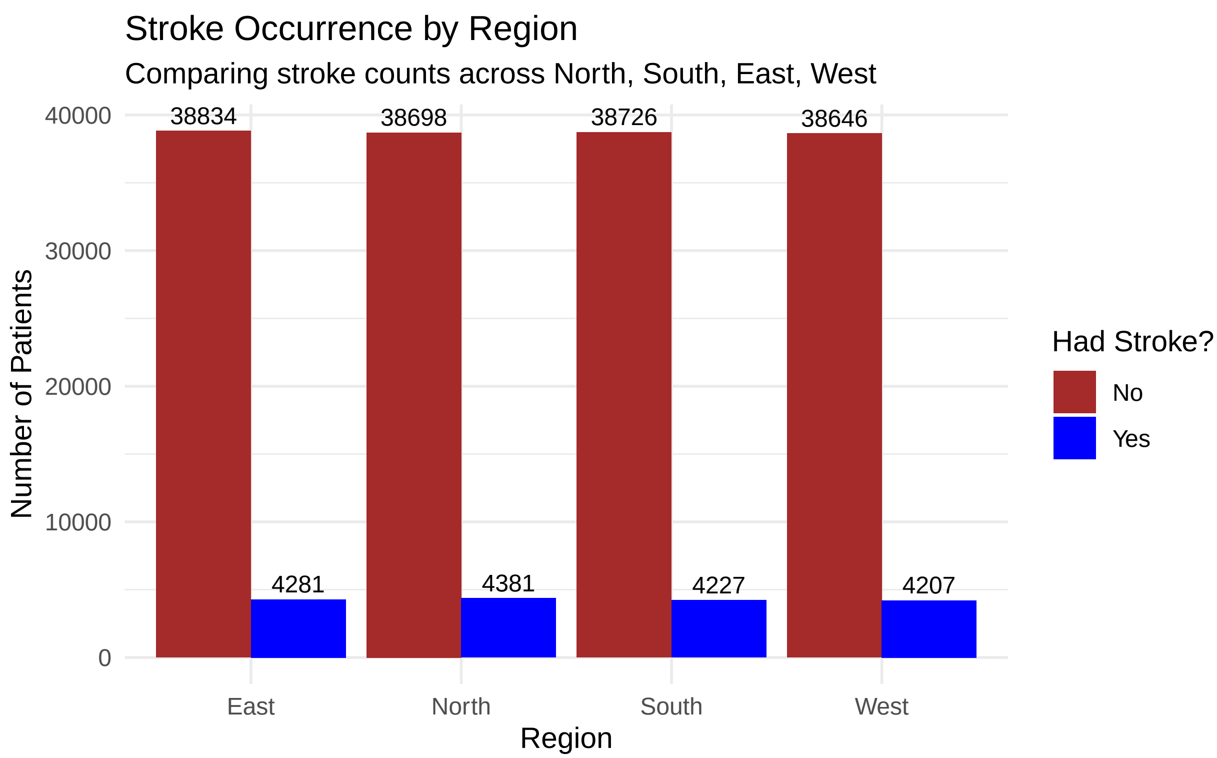

Figure 13: Stroke occurrence: Yes vs No by geographic region

6.5 Summary of Insights

Table 2: Summary of key trends and insights from the EDA

Insight	Finding
Age is the strongest predictor	Older patients consistently have higher stroke risk scores
Hypertension raises stroke risk	Patients with hypertension have more strokes than those without
Heart disease increases stroke risk	Both male and female patients with heart disease have more strokes
Smoking is associated with stroke	Smokers and former smokers have higher stroke counts in both genders
Sedentary lifestyle is risky	Sedentary patients have the highest stroke counts in the dataset
High glucose links to high risk	Higher glucose levels are positively correlated with stroke risk score
Gender differences exist	Patterns of heart disease and smoking risk appear in both male and female groups
Data quality is good	No missing values and no duplicates — the dataset is clean and complete

7 Conclusion

This Exploratory Data Analysis of the patient health dataset explored **171,999 patient records** across **22 features**.

Main conclusions:

1. **Age** is the strongest individual predictor of stroke risk — confirmed by histograms, boxplots, scatter plots, and the correlation matrix
2. **Hypertension** significantly increases the likelihood of stroke
3. **Heart disease** is associated with higher stroke counts in both male and female patients
4. **Smoking** is linked to higher stroke occurrence across both genders — active smokers and former smokers both show elevated stroke counts
5. **Sedentary patients** have the highest stroke counts — physical activity appears to be a protective factor
6. The dataset is **clean and complete** — no missing values or duplicates were found, making it reliable for further modelling

---

No Artificial Intelligence tools were used in the creation of this report. All work has been independently completed.